



TABLE OF CONTENT

Sr. No.	Experiment Title	Page No		Date of Performance	Date of Submission	Marks	Sign.
		To	From				
1.	Introduction to PLC	11	1	12/06/26	19/06/26	9	Ben 17.7
2	PLC program to implement the binary to gray code converter.	17	12	19/06/26	26/06/26	9	Ben 17.7
3.	PLC program to implement the binary to gray code converter.	21	18	26/06/26	03/07/26	9	Ben 17.7
4.	PLC program to implement 8x1 multiplexers using ladder logic diagram.	27	22	03/06/26	10/07/26	9	Ben 17.7
5.	PLC program to implement 1x8 demultiplexer using ladder logic diagram.	31	28	10/07/26	17/07/26	10	Ben 17.7
6.	PLC program to implement 8x3	33		17/07/26	07/08/26	10	Ben 7/8



TABLE OF CONTENT

Sr. No.	Experiment Title	Page No		Date of Performance	Date of Submission	Marks	Sign.
		To	From				
	ent the given logic function						
07	PLC program to implement Timers using ladder diagram	41	36	07/08/26	14/08/26	9	14/08
08	PLC program to implement counter using ladder logic diagram	47	42	07/08/26	14/08/26	10	14/08
09	PLC program to implement to control level of two tanks	51	48	14/08/26	11/09/26	10	11/09
10	PLC program to create car parking system	55	52	11/09/26	15/09/26	9	15/09



Experiment no. - 01.

Aim:- Introduction to programmable logic controllers (PLC).

Apparatus:- Rockwell Automation software for Allen Bradley PLC.

Theory:-

A programmable logic controller (PLC) is a digital, microprocessor-based industrial computer designed to control automated processes and machinery in real time. It continuously monitors input devices (sensors, switches, etc.) makes decisions based on a stored program, and controls output devices (motors, valves, relays, actuators) accordingly.

(2) Historical Background

- PLCs were developed in the late 1960s (Modicon, 1968) to replace complex relay-based control panels used in the automobile industry (General Motors).
- Relay logic required extensive rewiring for every change in control sequence; PLCs solved this by allowing logic changes through reprogramming rather than rewiring.

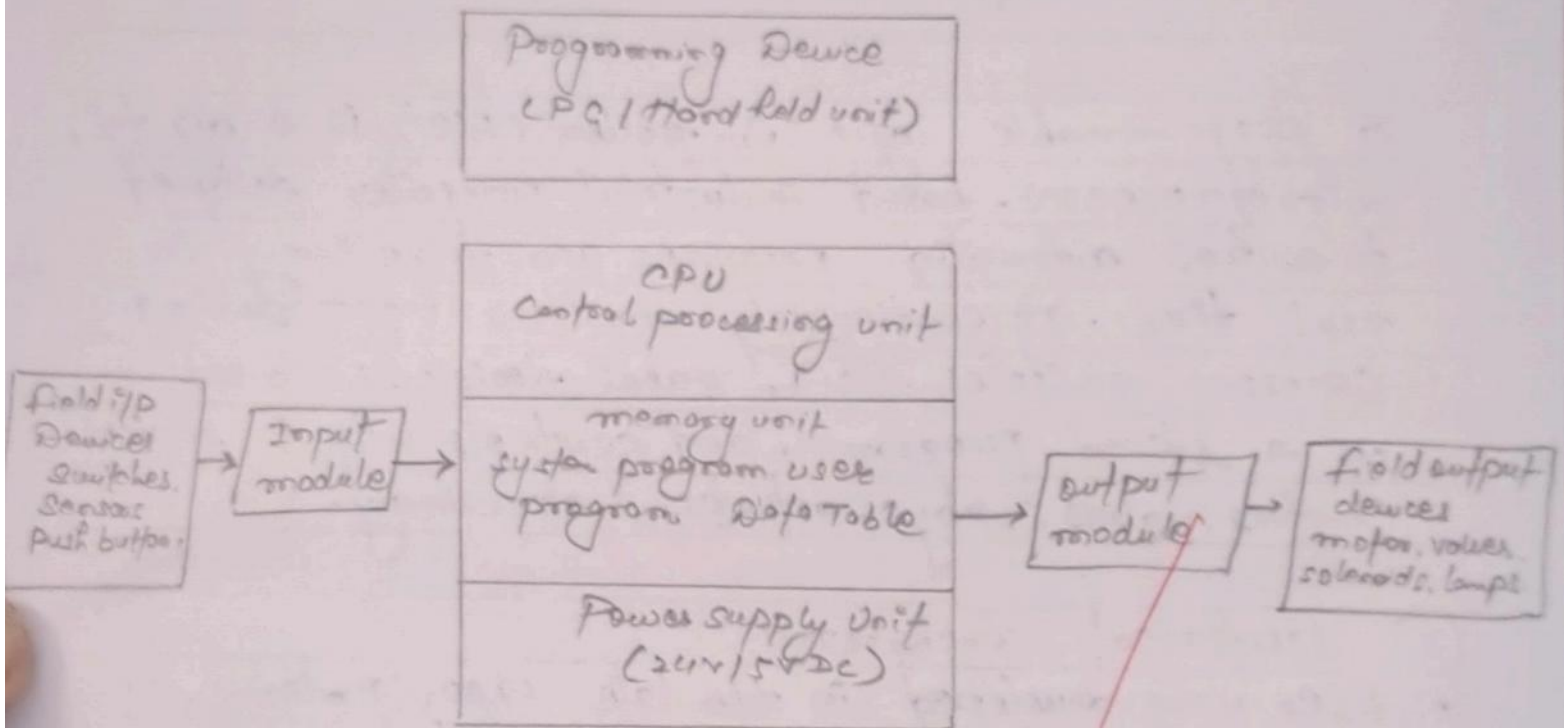


Fig.: Block Diagram / Architecture of a PIC





③ Need for PLC (Why PLC over Relay Logic?)

Relay Based Control

- Bulky panels, large space required
- Rewiring needed for logic change.
- Difficult troubleshooting.
- Poor reliability.
- Limited to simple sequential logic
- High maintenance cost
- No communication capability

PLC - Based control

- Compact, space-saving
- Reprogramming only, no rewiring.
- Build-in diagnostics, LED indication.
- High reliability, solid-state
- Handles complex logic, timers, counters, PID, arithmetic.
- Low maintenance
- Supports networking / SCADA / HMI integration.

④ Basic Block Diagram / Architecture of PLC

→ A PLC system consists of five main functional blocks:

1. Input module - Interfaces field input devices (push buttons, limit switches, proximity sensors, photoelectric sensors) with the PLC. Converts field signal levels to logic levels the CPU can process.

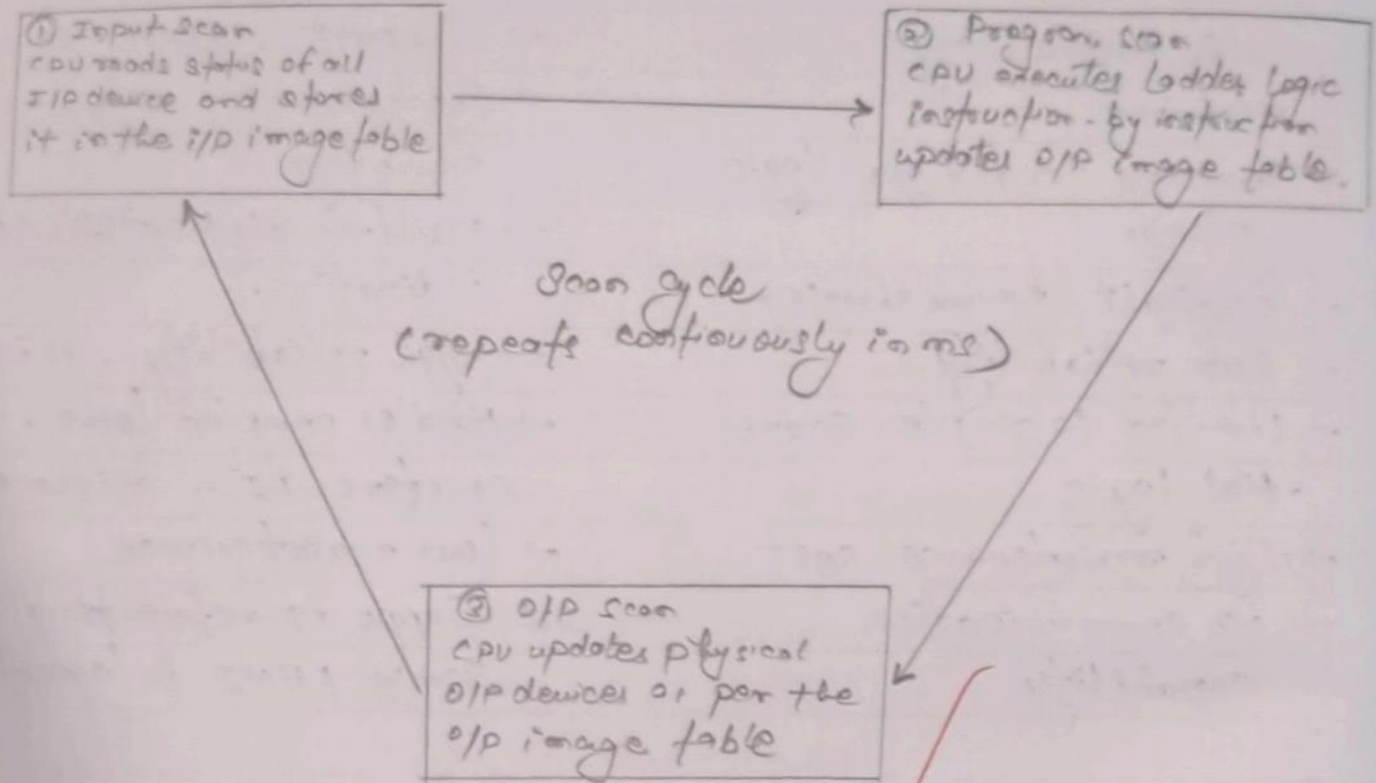


fig - PIC scan cycle.



2. CPU (Central processing Unit) - The "brain" of the PLC. Executes the user program stored in memory, performs logical and arithmetic operations, and controls overall PLC operation.
 3. Memory Unit - Stores the system programs (operating system supplied by manufacturer), the user program (adder logic written by the programmer), and the data table (status of Inputs/Outputs, timers, counters, internal relays).
 4. Output module - Converts CPU logic-level signals into power level signals to drive field output devices (motors, solenoids, lamps, contactors).
 5. Programming Device - A computer or handheld unit used to write, edit, download and monitor the ladder logic program (e.g. RS Logic, TIA portal, GX works, CX-programmer).
- Power Supply unit also forms part of the PLC, providing regulated DC voltage (typically 24V DC/5VDC) to internal circuitry.

Working Principle - The Scan cycle.

The PLC operates in a continuous, repetitive cycle called the scan cycle, consisting of three main stages:

- **Input Scan** - The CPU reads the status of all input devices and stores their ON/OFF states in the input image table.
- **Program Scan (Logic Execution)** - The CPU executes the ladder logic program instruction-by-instruction, using the input image table, and updates the output image table based on the program logic.
- **Output Scan** - The CPU updates the actual physical output devices according to the output image table. This cycle repeats continuously (typically in milliseconds) and is called scan time.

Types of PLC (Based on structure)

- ① **Fixed / Compact PLC** - I/O, CPU, memory and power supply are integrated into a single housing (e.g. - small automation tasks).
- ② **Modular PLC** - CPU, I/O modules and power supply are separate, mounted on a rack. I/O count can be expanded as required (used for large industrial applications).

Classification Based on I/O capacity.

Type	No. of I/O.
micro PLC	upto 32 I/O
Small PLC	32-128 I/O

medium PLC

128-512 I/O

Large PLC

above 512 I/O.

Application of PLC.

- Automobile assembly lines
- Material handling and conveyor systems.
- Packaging machines
- Elevators and traffic light control
- Water treatment plants.
- Process industries (chemical, textiles, cement, food processing)
- SCADA-integrated systems for remote monitoring and control.

Advantages of PLC

- Flexible - logic changed via software, not hardware.
- Compact and reliable (solid-state design)
- Fast response time.
- Easy troubleshooting with diagnostic indicators.
- Capable of complex operations: timing, counting, arithmetic, PID control.
- Supports communication protocols (modbus, profibus, Ethernet/IP) for SCADA/HMI integration.

Leading PLC manufacturers:

- Allen-Bradley (Rockwell Automation)
- Siemens
- Schneider electric (modicon)
- Mitsubishi
- Omron
- ABB
- Delta

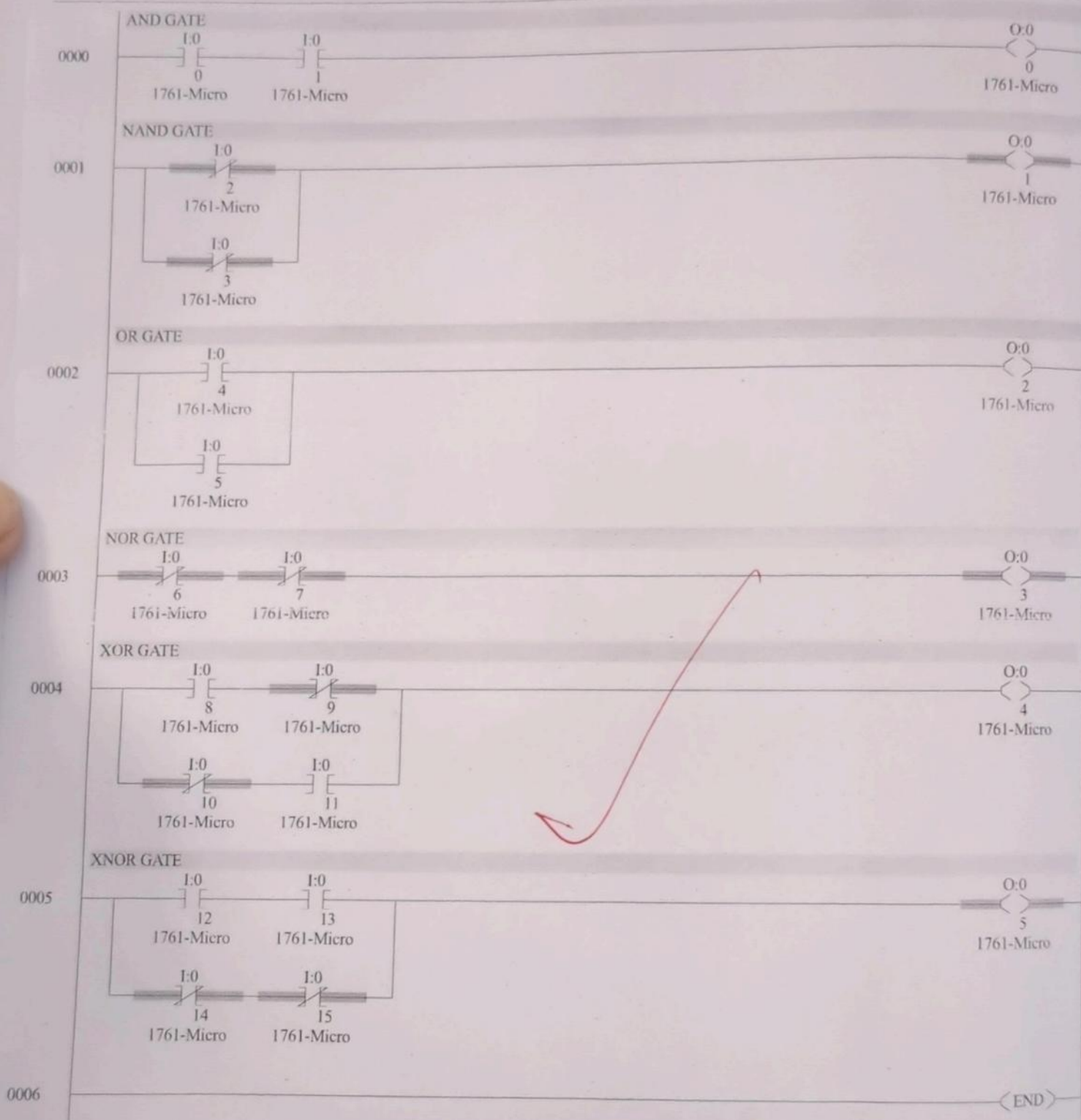
Conclusion:- The experiment helped us understand the basic working components and applications of PLC. It provides a foundation for PLC programming and industrial automation.

✓
Rev
17.7.26



EXP 2 - LOGIC GATES

LAD 2 - --- Total Rungs in File = 7



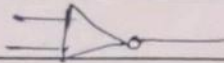
Experiment no. - 02

Aim: PLC program to implement various digital logic gates using Ladder Logic diagram.

Apparatus : RS logic 500

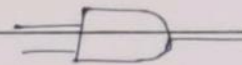
Theory: Logic gates are electronic circuits that perform logical functions. These gates process binary inputs (0 & 1) and produce binary outputs based on predefined logical operations. The logic gates and their truth table are

NOT



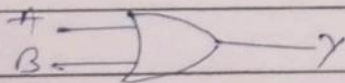
Input	Output
A	Y
0	1
1	0

AND

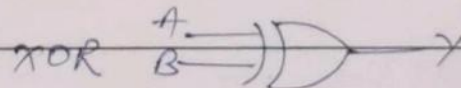


Input	Output
A B	Y
0 0	0
0 1	0
1 0	0
1 1	1

OR



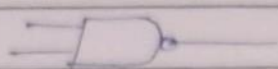
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1



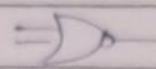
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0



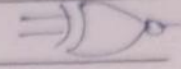
NAND



NOR



XNOR



Input			Output	Input			Output	Input			Output
A	B		Y	A	B		Y	A	B		Y
0	0		1	0	0		1	0	0		1
0	1		1	0	1		0	0	1		0
1	0		1	1	0		0	1	0		0
1	1		0	1	1		0	1	1		1

Ladder programming

Ladder diagram is a popular language of programming the PLC. Ladder diagram shows the sequence of logic execution which is presented diagrammatically. In ladder diagram, there are two vertical lines generally called as phase (positive) and neutral. Rungs which shows the current flow in horizontal direction are the sequence in which the logic executes. The analogue to relay ladder has two main symbols which are contacts and output coil.

Generally each rung has input on the left hand and output coils on the right side. These contacts and coils are called a bit of the relays. Each input and output are individual bit in I/O files. An instruction in ladder instructs PLCs how to respond to this

which are stored in the memory. Input contacts are the condition area & the conditions area must be fulfilled to change the states of the output coils.

Some commonly used relay instructions:

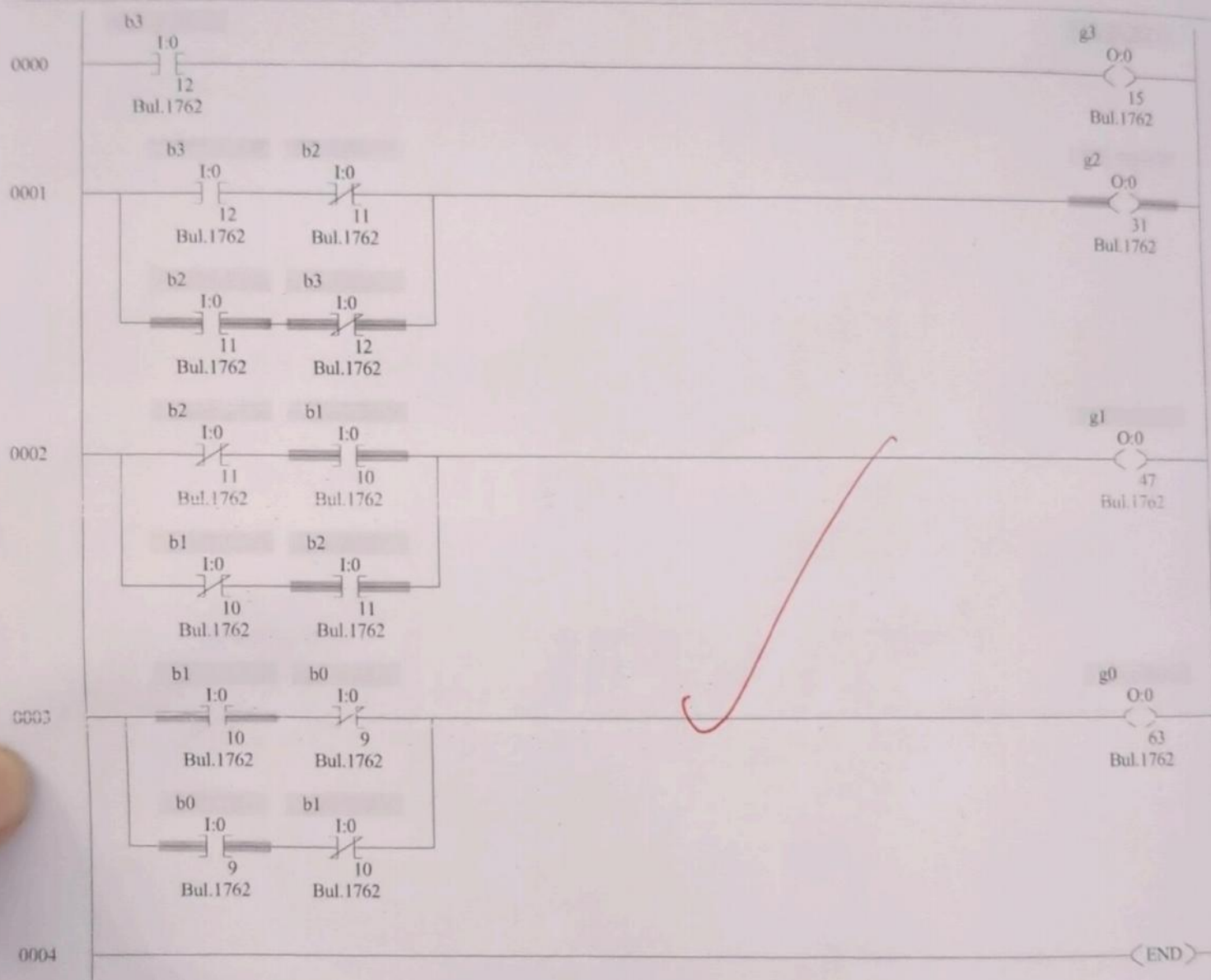
Instruction	Symbol	Description
• Normally open switch	— —	→ An input condition that is opened which de-energized.
• Normally closed switch	—/ —	→ An input condition is close when de-energized.
• Output coil	—()—	→ An o/p condition i.e true all the time except when all the I/P condition are true.

Conclusion: By performing this experiment we successfully digital logic gates using ladder logic. It demonstrates the basic logic function used in industrial automation.

17.7

BINARY TO GRAY CONVERSION

LAD 2 - --- Total Rungs in File = 5





Experiment no-03.

Aim: PLC program to implement the binary to gray code converter.

Apparatus: RS Logic 500

Theory: Binary no. is a difficult way to store no., but in many applications, binary no. are difficult to use & a variety of binary nos. is needed. This is where Gray codes are very useful.

Gray codes are very useful. Has a property that two successive nos. are differ in only one bit because of this property gray code does the cycling through various states with minimal effort and is used in K-maps, error correction, communication etc.

Steps to convert the binary code to gray code:

- The MSB (most significant Bit) of the gray code will be exactly equal to the first bit of the given binary no.
 - The second bit of the code will be exclusive OR (XOR) of first and second bit of the binary no. i.e. both the bits are same, the result will be zero and if the both bit are different then result will be 1.
 - The third bit of gray code will be equal to the XOR of second and third bit of the given binary no.
- Thus the binary to gray code conversion goes on. An example



Binary				Gray code			
B_3	B_2	B_1	B_0	G_3	G_2	G_1	G_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

B_3, B_2	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

$$G_3 = B_3$$

B_3, B_2	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	0	0	0	0
10	1	1	1	1

$$G_2 = B_2 \oplus B_3$$

B_3, B_2	00	01	11	10
00	0	0	1	1
01	1	1	0	0
11	1	1	0	0
10	0	0	1	1

$$G_1 = B_2 \oplus B_1$$

B_3, B_2	00	01	11	10
00	0	1	0	1
01	0	1	0	1
11	0	1	0	1
10	0	1	0	1

$$G_0 = B_1 \oplus B_0$$

therefore the eqn is

$$\rightarrow G_3 = B_3$$

$$G_2 = B_3 \oplus B_2$$

$$G_1 = B_2 \oplus B_1$$

$$G_0 = B_1 \oplus B_0$$





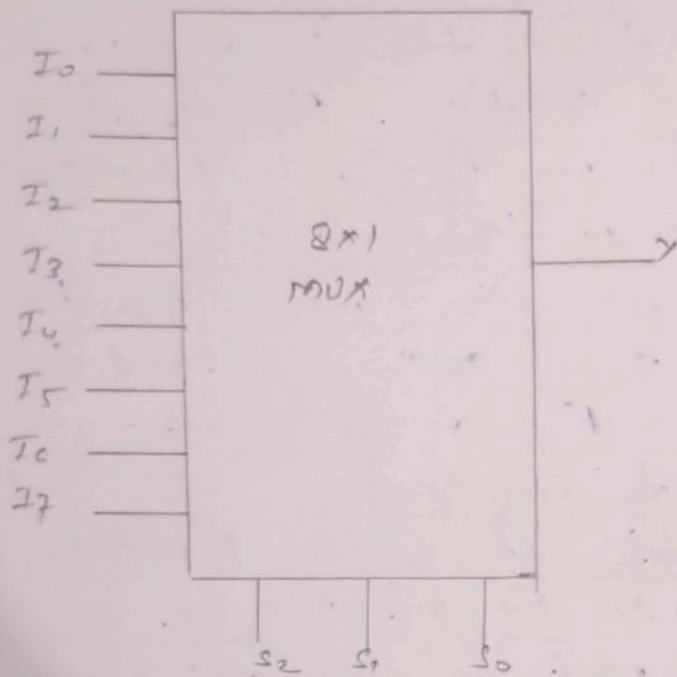
is given below to illustrate these steps.

Some of the application of gray code:

- Gray code can be used to simplify Boolean function in Karnaugh's maps. The advantage comes from the fact that this allows you to fill the Karnaugh's maps and make groupings in them. Thereby, it reduces the complexity and size of the original eqn.
- This code is used mainly in position systems either angular or linear. Its main applications are in industry and robotics. In Robotics, coded discs are used to give the position information that an axis has. This information is given to gray code.
- In communication systems, it is used to detect unexpected changes in data. If the bits in a no. are summed, the sum of the next no. should only change by one with the sum alternating even and odd.

Conclusion: In this experiment the PLC program for Binary to Gray code conversion was implemented successfully. The experiment improved our understanding of PLC programming, logic operations, and code conversion techniques.

17.7



I/P's	S_2	S_1	S_0	Y (O/P)
I_0	0	0	0	I_0
I_1	0	0	1	I_1
I_2	0	1	0	I_2
I_3	0	1	1	I_3
I_4	1	0	0	I_4
I_5	1	0	1	I_5
I_6	1	1	0	I_6
I_7	1	1	1	I_7





Experiment no.-011,

Aim: PIC program to implement 8×1 multiplexer, using Ladder logic diagram.

Apparatus : RS logic 500

Theory: A multiplexer, often abbreviated as MUX, is a digital circuit that allows multiple input signals to be transmitted over a single channel or line. It is commonly used in digital electronics and communication systems. The primary function of a multiplexer is to select one of input signals and route it to a single output line. The selection is controlled by a set of control inputs, typically called select lines. The no. of select lines determines the no. of input lines that can be accommodated by the multiplexer. A multiplexer with n select lines can handle 2^n input lines.

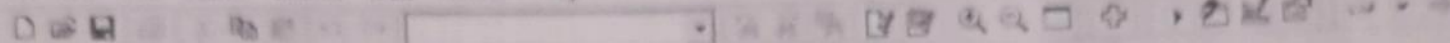
Multiplexers are often represented using a block diagram or a truth table. The block diagram shows the inputs, select lines and output line. The truth table specifies the output value for each combination of input and select line states.

8×1 MUX :

An 8 to 1 multiplexer (8×1 MUX) is a multiplexer that

RSLogix 500 Prog - MLX

File Edit View Search Commits Tools Window Help



REMOTE RUN

No Forces

Forces Enabled

No Edit

Forces Enabled

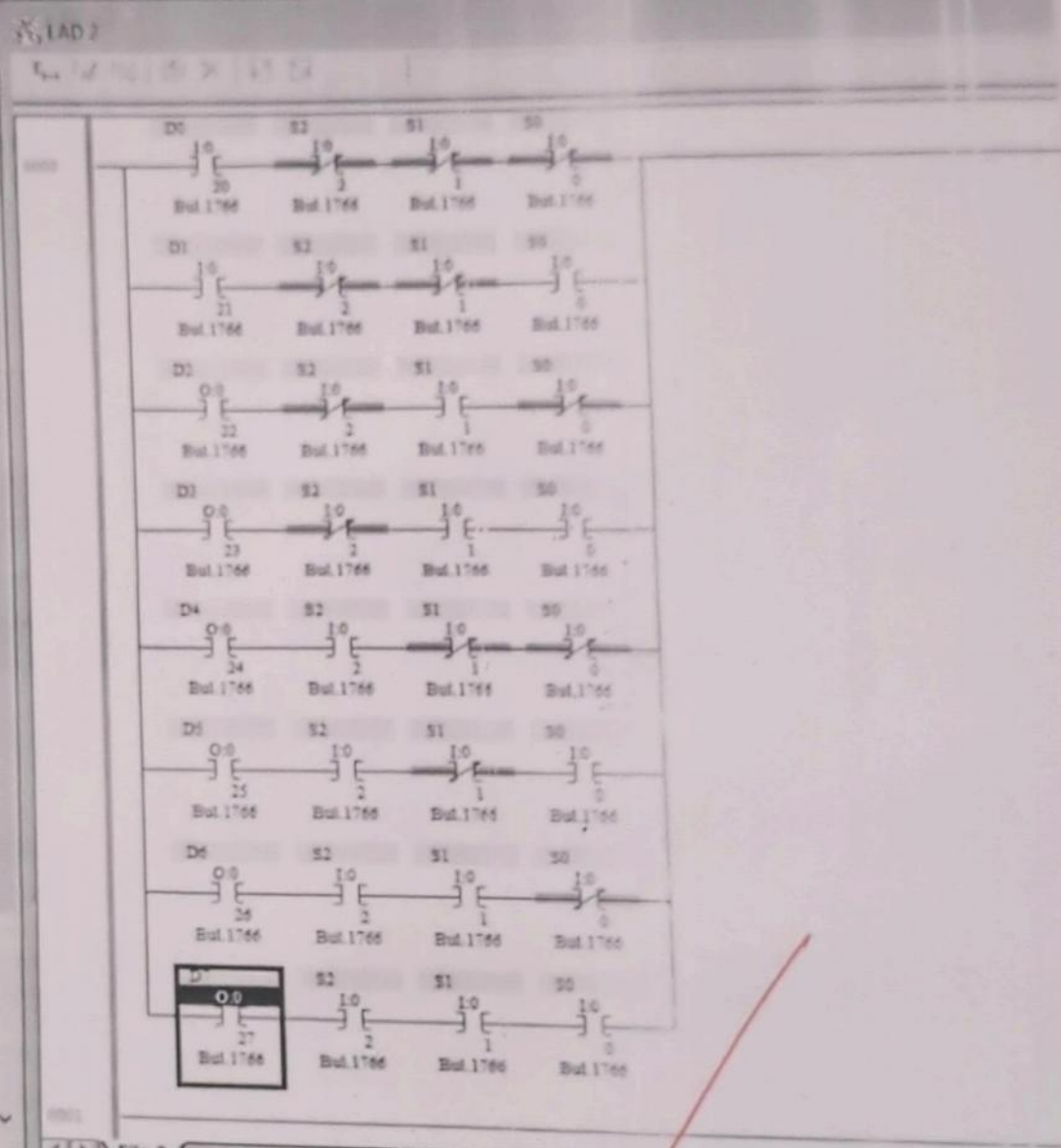
Driver EMU500-1

Node 1d

User Bit TimerCounter InputOutput Compare

Project Explorer

- Project
 - Help
 - Controller
 - Controller Properties
 - Processor Status
 - Function Files
 - ID Configuration
 - Channel Configuration
 - Program Files
 - SYS 0 -
 - SYS 1 -
 - LAD 2 -
 - Data Files
 - Cross Reference
 - O0 - OUTPUT
 - I1 - INPUT
 - S2 - STATUS
 - B3 - BINARY
 - T4 - TIMER
 - C5 - COUNTER
 - R6 - CONTROL
 - N7 - INTEGER
 - F8 - FLOAT
 - Data Logging
 - Configuration
 - Status
 - RCP Configuration Files
 - Force Files
 - O0 - OUTPUT
 - I1 - INPUT
 - Custom Data Monitors
 - CDM 0 - Untitled
 - Custom Graphical Monitors
 - CGM 0 - Untitled
 - Recipe Monitors
 - Recipe 0 - Untitled



Valid Entry created in Database!

has 8 input lines and 1 output line. It selects one of the 8 I/P's lines and routes its value to the o/p based on the select I/P's. In the block diagram of mux I_0 to I_7 represents the 8 I/P's lines and y represents the output line. S_0 and S_1 are the select lines that controls the inputs selection. The combination of S_0 and S_1 determines which input line is connected to the output. The truth table for 8x1 mux would have rows, each representing the state of the select inputs (S_0, S_1) and corresponding output value (y) based on the selected input lines.

for example,

If $S_0 = 0$ and $S_1 = 1$, the input I_2 will be selected and its value will appear on the output line y .

Multiplexers with different nos. of input and select lines are commonly used in digital systems to efficiently route data and control signals.

the output equation for 8x1 multiplexer

$$\begin{aligned}
 Y_0 = & S_0 S_1 S_2 I_0 + S_0 S_1 \bar{S}_2 I_1 + S_0 \bar{S}_1 S_2 I_2 + S_0 \bar{S}_1 \bar{S}_2 I_3 + \bar{S}_0 S_1 S_2 I_4 \\
 & + \bar{S}_0 S_1 \bar{S}_2 I_5 + \bar{S}_0 \bar{S}_1 S_2 I_6 + \bar{S}_0 \bar{S}_1 \bar{S}_2 I_7
 \end{aligned}$$



Observation: From the above experiment we have observed the following observations in 8x1 multiplexer using ladder

Inputs signals: The experiment would involve eight input signals (I_0 to I_7) and one select lines (S_0). These signals are usually represented by switches or push buttons in the ladder diagram.

Select signal: The Select signal (S_0) determines which input signal gets routed to the output. It is usually connected to a switch or push button and can be manually controlled.

Output signal: The output signal (Y) represents the selected inputs signals based on the value of the select signal. The output is usually connected to an indicator, such as an LED, to visually display the result.

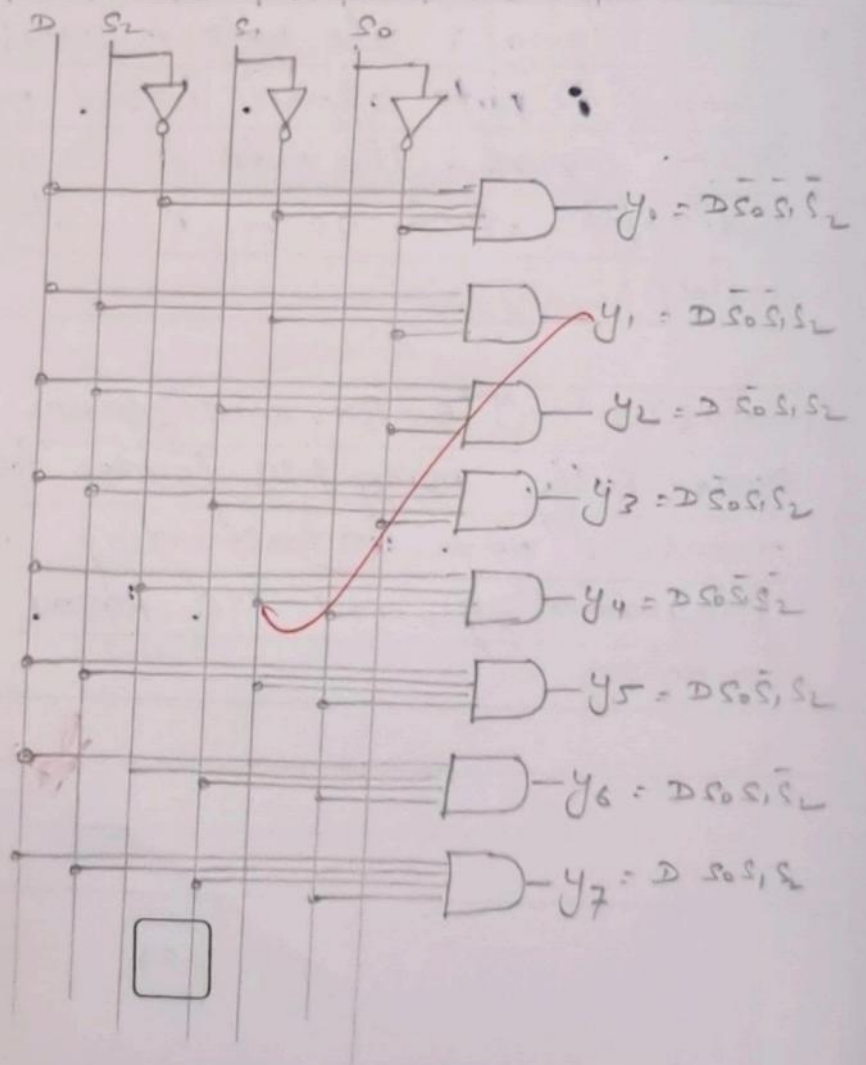
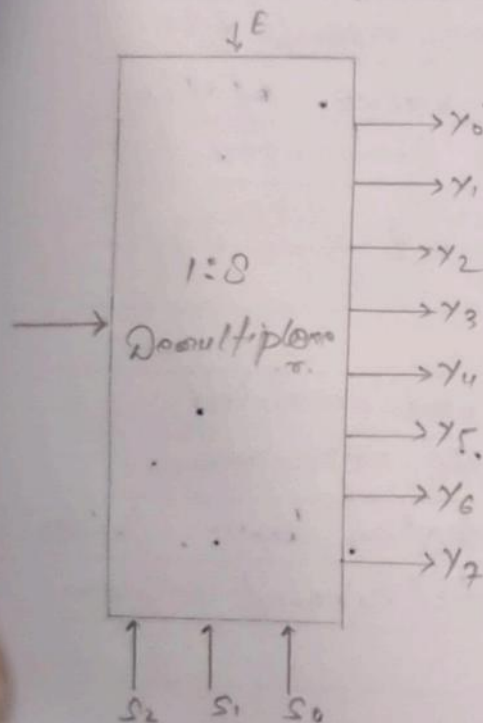
Conclusion: The 8x1 multiplexer was implemented successfully using PLC ladder logic. The experiment improved our understanding of selection logic, ladder programming, and PLC based digital circuit implementation.

Rev
17.7



→ 1-to-8 demultiplexer truth table and circuit diagram

Data i/Ps	Select i/Ps										
D	S ₂	S ₁	S ₀								
D	0	0	0	0	0	0	0	0	0	0	D
D	0	0	1	0	0	0	0	0	0	D	0
D	0	1	0	0	0	0	0	0	D	0	0
D	0	1	1	0	0	0	0	D	0	0	0
D	1	0	0	0	0	0	D	0	0	0	0
D	1	0	1	0	0	D	0	0	0	0	0
D	1	1	0	0	D	0	0	0	0	0	0
D	1	1	1	D	0	0	0	0	0	0	0



Experiment no. 05

Ques:- PIC program to implement 1:8 demultiplexer using ladder logic diagram.

Apparatus : RS Logic 500

Theory: A demultiplexer (also known as demux or data distributor) is defined as a circuit that can distribute or deliver multiple outputs from a single input. A demultiplexer can perform as a single input with many output switches. The demultiplexer output line are n in number, the select line no. is m and $n = 2^m$. The control signal or select input code decides the output line to which the input has to be transmitted.

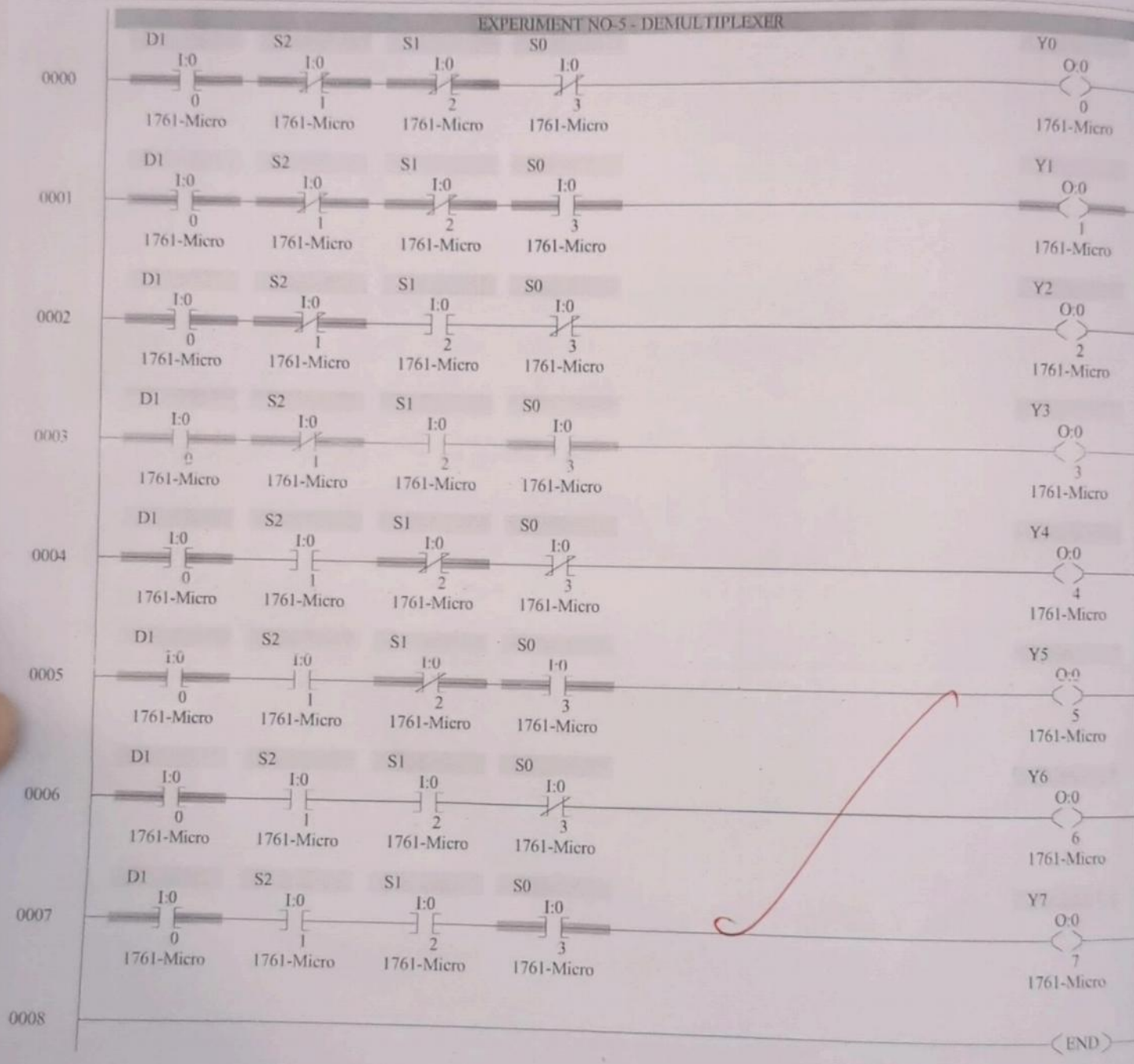
The demux can also perform as a binary to decimal decoder. The data input line should be at the logic level, and the binary input is given to the select line. The corresponding line will be given the output. In the designing of multiple combinational circuits, this circuit advantages.

1 to 8 demultiplexer performs the transfer of a single data to any one of the 8 possible outputs. It has 3 selection lines to distribute the data to the output.



PLC EXP 5 DEMULTIPLEXER

LAD 2 - MAIN_PROG --- Total Rungs in File = 9



Observations:

- **Input signals:** The experiment uses one input signal (I) and select lines (S_0, S_1, S_2) . These signals are usually represented by switches or push buttons in the ladder diagram.
- **Select lines:** The select lines (S_0, S_1, S_2) determine which output receives the input signal. They can be manually controlled using switches or push buttons.
- **Output signals:** The output signals $(Y_0 \text{ to } Y_7)$ represents the selected output. Based on the values of the select lines the input signal is routed to only one output, while all other outputs remain off. The outputs are usually connected to LED's for visual indication.

Conclusion: The PLC program for the demultiplexer was implemented successfully. The experiment improved our understanding of Ladder logic, signal routing and PLC-based digital circuit implementation.

Rev
17.7

Experiment no. -06

Ques: PLC program to implement the given logic function
 $f(B_3, B_2, B_1, B_0) = m(0, 2, 7, 8, 9, 10, 11)$
 $g(B_3, B_2, B_1, B_0) = m(1, 5, 6, 12, 13, 14) + d(4)$

Apparatus: RS logic 500

Theory: In order to find the characteristics equation from canonical form, we need to draw k-map then after grouping gives the final characteristic equation.

- for function 'f', the k-map and characteristics is as follows:

① $f(B_3, B_2, B_1, B_0) = \sum m(0, 2, 7, 8, 9, 10, 11)$

$B_3 B_2 \backslash B_1 B_0$		$B_1 B_0$			
		00	01	11	10
00	1	0	0	1	
01	0	0	1	0	
11	0	0	0	0	
10	1	1	1	1	

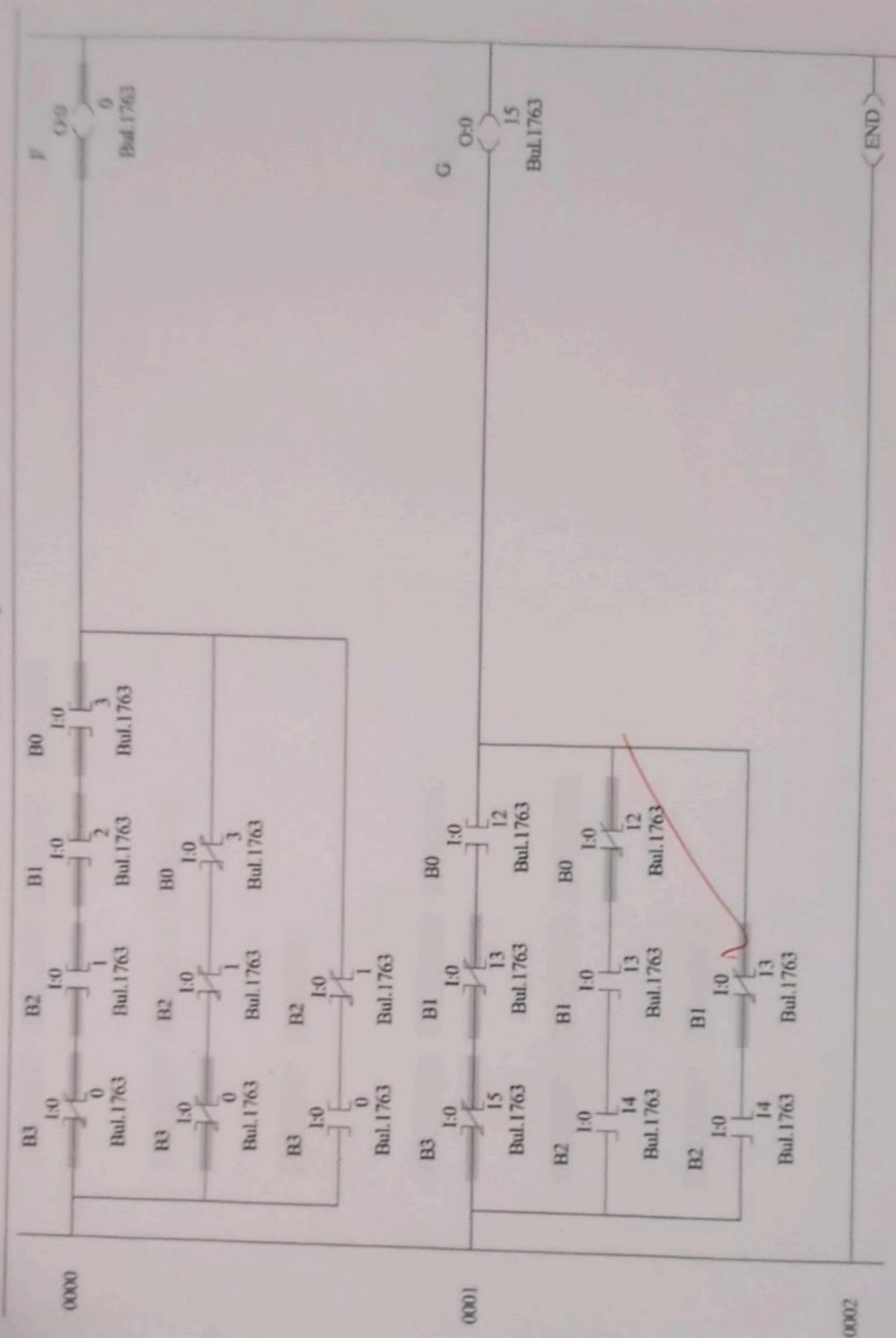
$$f(B_3, B_2, B_1, B_0) = B_3 B_2 + \bar{B}_2 B_0 + \bar{B}_3 B_2 B_1 B_0$$

- for function 'g' the k-map and characteristics eqn is as follows.



LOGIC FUNCTION IMPLEMENTATIONEXP6_BAK001

LAD 2 - --- Total Rungs in File = 3





$$② \quad g(B_3, B_2, B_1, B_0) = m(1, 5, 6, 12, 13, 14) + d(4)$$

$B_3 B_2 \backslash B_1 B_0$	00	01	11	10
00	0	1	0	0
01	x	1	0	1
11	1	1	0	1
10	0	0	0	0

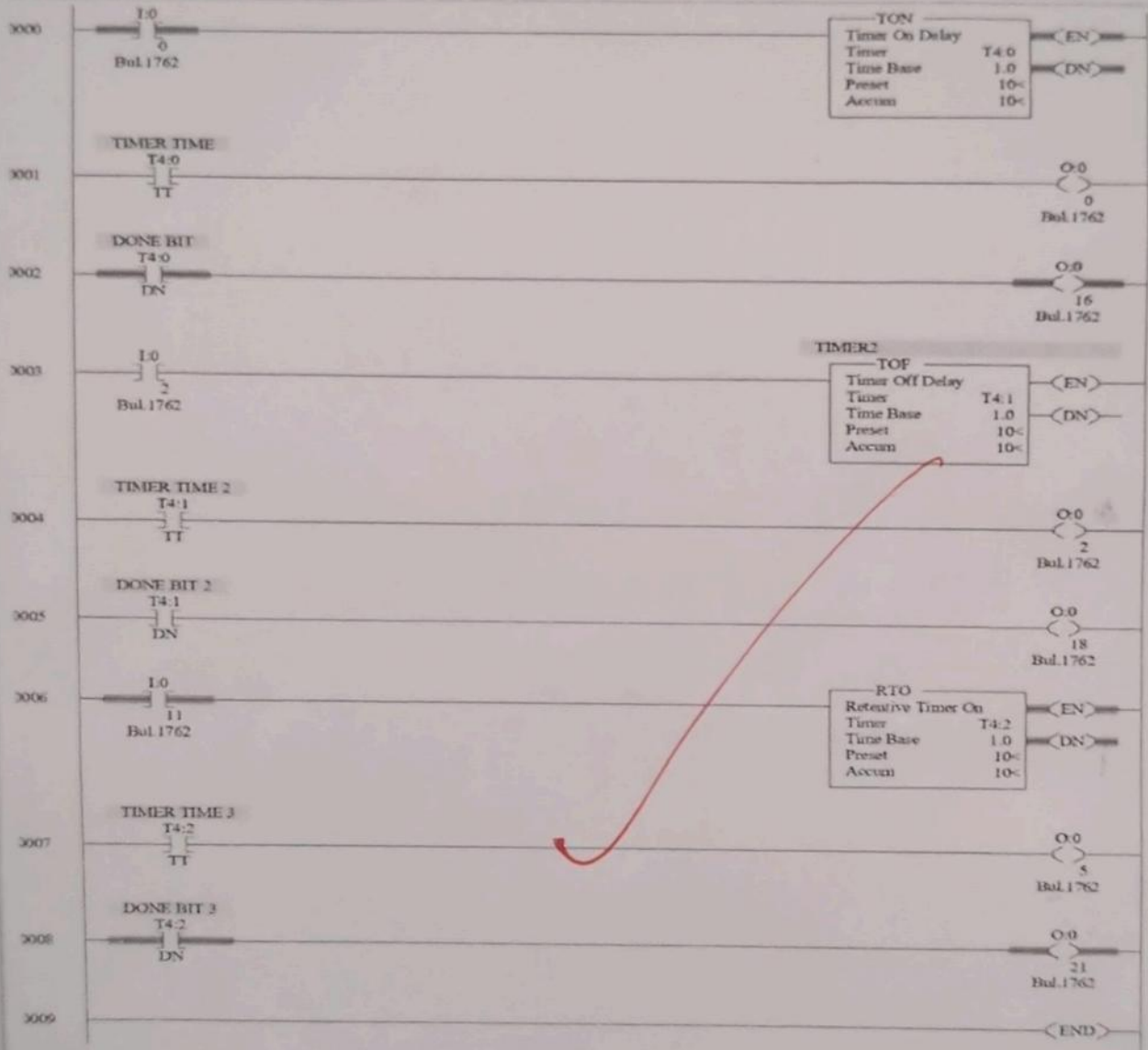
$$g(B_3, B_2, B_1, B_0) = B_2 \bar{B}_0 + B_2 \bar{B}_1 + \bar{B}_2 \bar{B}_1 B_0$$

Observation:

- Given are the minterms of a function 'F' whose characteristics equation can be obtained by simplifying from the k-map
- After obtaining the characteristics equation, we can implement the function simply in PLC online simulator.
- We cannot perform this in virtual labs, since there should be 3 compulsory loops in a single rung, which cannot be compiled in the virtual labs.
- We do the same for the other function to obtain the results.

Conclusion: In this experiment, the Boolean functions were successfully simplified using k-map and implemented via ladder logic diagram in a PLC simulator.

Testing across various input combinations matched the expected truth table, verifying the correct implementation of the simplified expressions.





Experiment no.-07.

Aim:- PLC program to implement Timers using Ladder diagram

Apparatus - RS Logic 500

Theory -

In the context of programmable logic controllers (PLCs), timers are essential for controlling processes and operations based on time intervals. There are some key points about timers in PLCs:

On Delay Timer (TON) :-

An ON delay timer, also known as a delay on timer, is a type of timer used in control systems and electrical circuits. It is designed to introduce a delay between the activation of an input signal and the activation of an output signal. In other words, after the input signal is turned on, there is a specific duration before the output signal is activated.

The ON delay timer is commonly used in various industrial applications and automatic systems where timing functions are required. It allows for controlling the duration of an operation or providing a delay before energizing a circuit or starting a process.

The working principle of an ON delay timer is as follows.

When the input signal is activated, the output signal remains inactive for a predefined period called the delay time. After this delay time elapses, the output signal is then activated. This delay allows for proper synchronization and coordination between different components or systems.

OFF Delay Timer (TOFF):

An OFF delay timer is a type of timer used in control systems and electrical circuits. It is designed to introduce a delay between the removal of an input signal and the deactivation of an output signal. In other words, after the input signal is turned off, the output signal remains active for a specific duration before it is switched off.

The off delay timer is commonly used in various industrial applications and automation systems where timing functions are required. It allows for controlling the duration of an operation or providing a delay before deenergizing a circuit or shutting down a process.

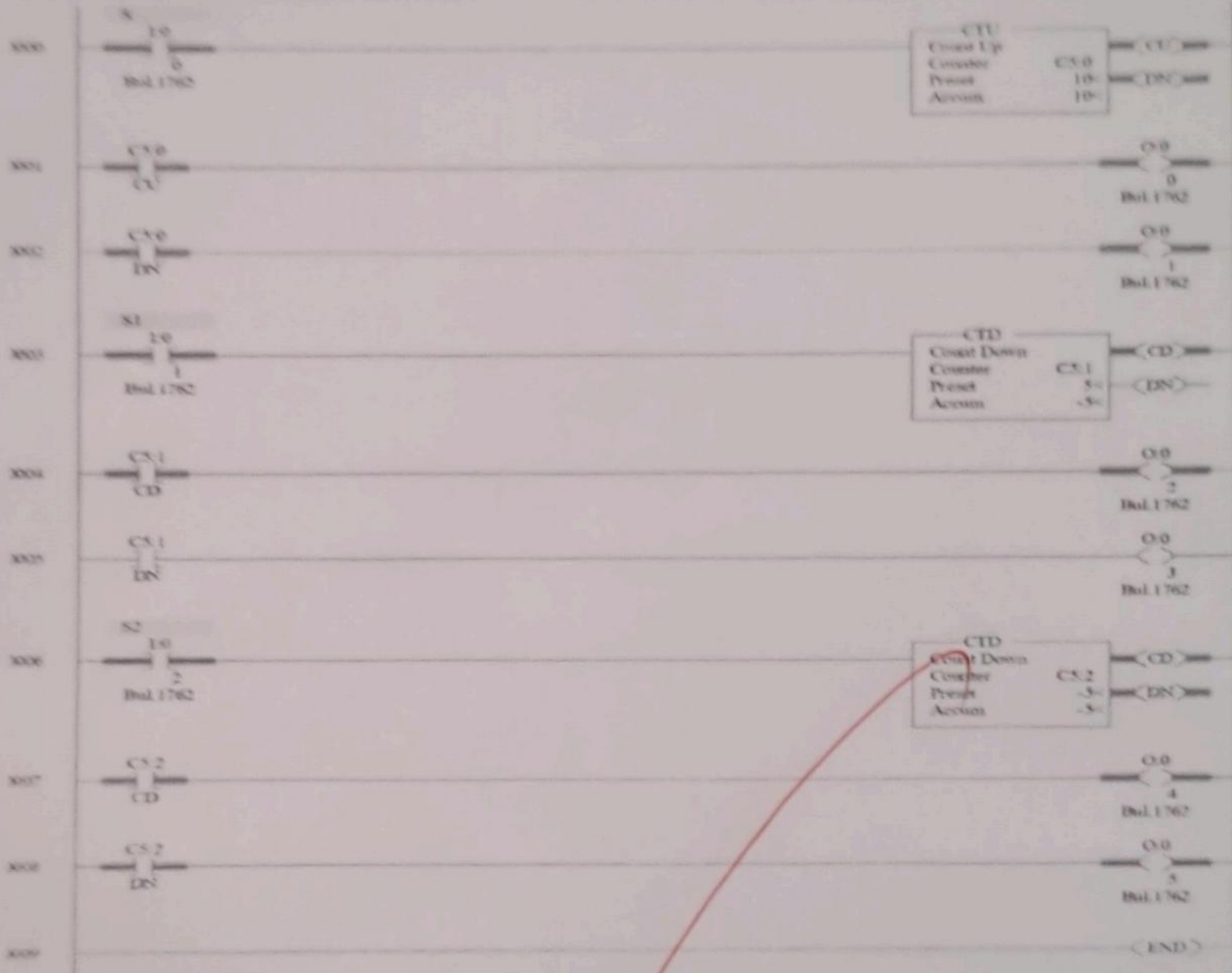
The working principle of an off delay timer is straightforward. When the input signal is activated, the output signal is also activated simultaneously. However, when the input signal is turned off, the output signal remains active for a predefined period called the delay time.



Conclusion :-

In this experiment, It showed that PLC timers can effectively control delays in automation. Using ON - delay and OFF delay timers in ladder logic, outputs responded accurately to inputs, confirming their importance in precise and reliable process control.

✓
R. R. 14/09





Experiment no.-08.

Aim: PLC program to implement counters using ladder logic diagram.

Apparatus : - PLC online simulator RS logic 500.

Theory -

A counter is a simple device intended to do one simple thing - count. Every PLC has counter instructions. Using counter sometimes be little challenging because many manufacturers seem to use them different way. In other words, the instruction symbol used and method of programming will change for different manufacturers.

Up Counters:

In programmable logic controllers (PLCs), up counters are essential for tracking the number of events or items that occur over time. An up counter in a PLC increments its count each time it receives a triggering signal, such as a clock pulse or a specific input signal, allowing for easy monitoring of processes like production counts or machine cycles.

Typically, PLC up counters have features such as preset values, which allows users to set a maximum count before the counter resets or rolls over to zero. They can also include reset functions that return the counter to its initial state.



The use of up counters in PLC applications is widespread, from counting products on an assembly line to monitoring operational cycles in machinery. Understanding how to implement and use up counters effectively enhances automation capabilities and improves the overall efficiency of industrial processes.

Down Counters:

In programmable logic controllers (PLCs), down counters are used to decrement a count based on specific conditions or events. Unlike up counters, which increment the count, down counters reduce their value with each triggering signal, allowing for applications such as countdown timers or tracking remaining items in a process.

A down counter typically starts at a preset value and counts down to zero. It can be triggered by various inputs, such as a timer or specific sensor signals. Once it reaches zero, the counter can either stop counting or reset to the preset value, depending on its configuration.

Down counters are useful in applications like inventory management, where they can track the number of items left in stock, or in production processes where they help manage remaining cycles or tasks.



Conclusion -

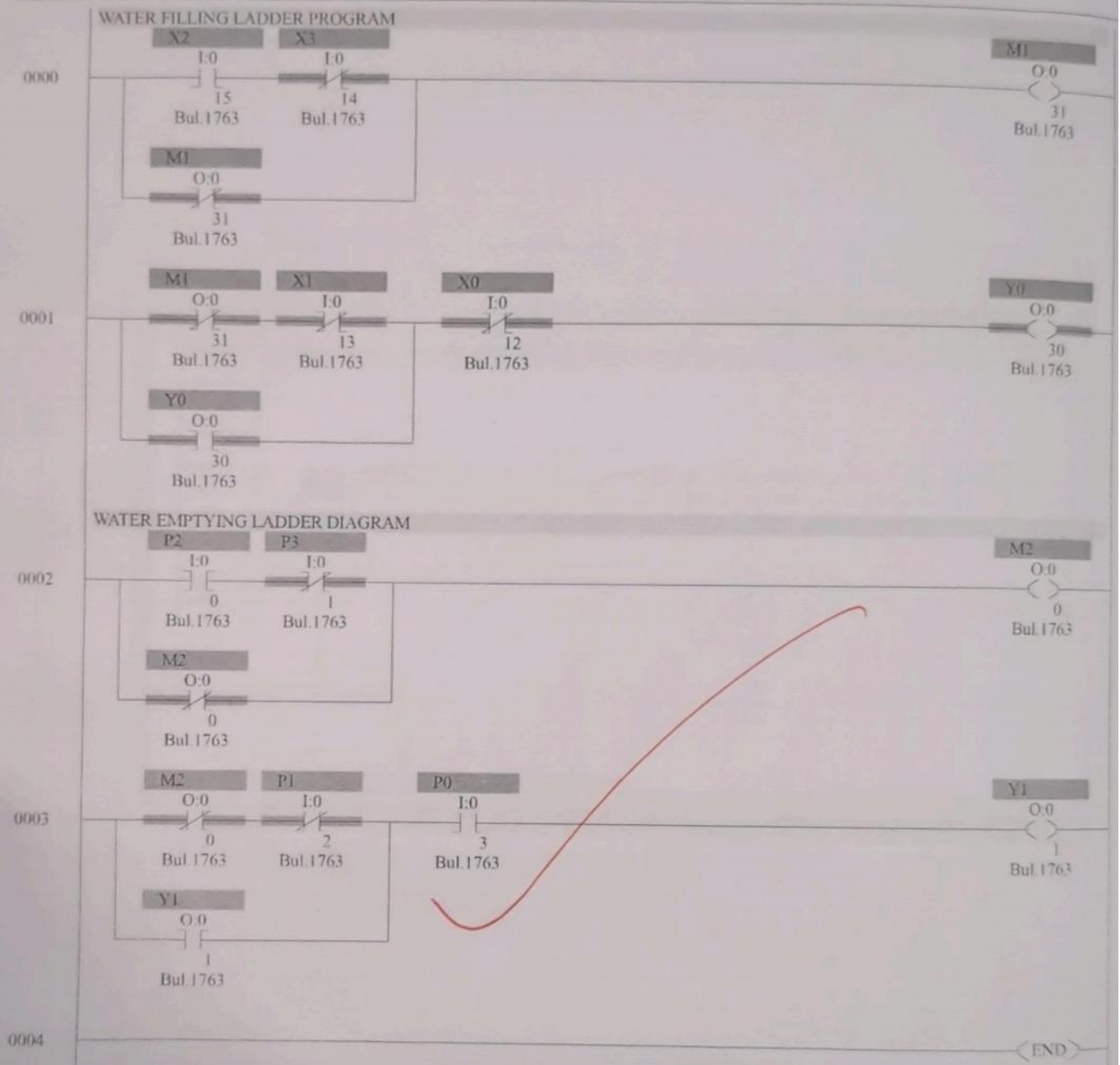
In this experiment, a PLC program was successfully designed and verified to implement counter operations using a ladder logic diagram. The system accurately tracked input pulses to update the accumulator value until it met the preset limit, seamlessly triggering the output via the Done bit. Ultimately, the experiment proves the PLC based counters are reliable, efficient and essential tools for automated manufacturing and industrial process control.

Rev
14/08



WATER LEVEL CONTROL SYSTEM_BAK001

LAD 2 - --- Total Rungs in File = 5





Experiment no. 09.

Aim. To study and implement the automatic water level control system using PLC for controlling the filling and emptying of a water tank.

Apparatus. RS Logix 500.

Theory -

Water level control is an important application of PLC used to automatically maintain the water level in a tank.

In the filling operation, when the water level falls below the low level sensor, the PLC switches ON the water pump and the tank starts filling. The pump continues to operate until the water reaches the high-level sensor, after which the PLC switches off the pump to prevent overflow.

In the emptying ~~pump~~ operation, the ~~water~~ reaches the high level sensor, the PLC switches ON the emptying pump or valve. Water is then removed from the tank until it reaches the low level sensor, at which point the PLC switches off the pump or valve. Thus, the PLC controls the water level automatically according to the signals received from the level sensors.

Observation -

During the experiment, it was observed that the water pump automatically started when the water level reached the low level condition during filling and stopped when the water reached the high level condition. During emptying, the pump started when the water reached the high level condition and stopped when the water level reached the low level condition. Hence, the PLC successfully controlled the water level between the required minimum and maximum levels.

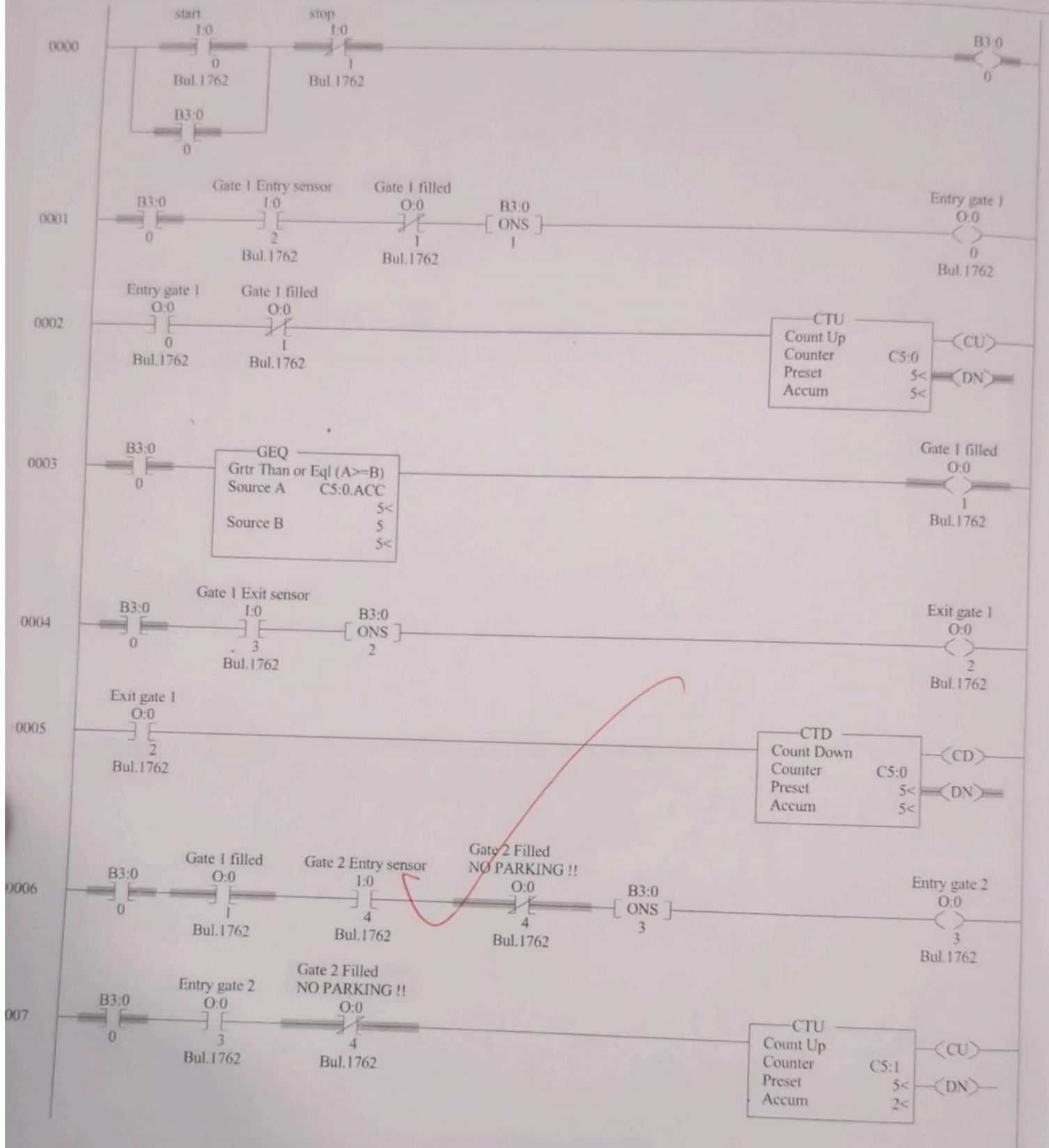
Conclusion -

In this experiment, the water level control system was successfully implemented using PLC. The filling and emptying of the tank were automatically controlled according to the water level sensors.

10/11/09

CAR PARKING HARSHA

LAD 2 - --- Total Rungs in File = 12





Experiment no. - 10

Aim. PLC Program to execute Car Parking System.

Apparatus - RS Logic 500

Theory -

We are asked to perform a parking system in Ladder Diagram. Assume there is a barricade at both Entry and Exit, such that it will open when car wants to come in or go out, the barricade will open at Entry and Exit respectively. I have programmed the Ladder Diagram in such a way that the barricades will not open when parking area is full. I assumed that

- The strength of parking lot is 10.
- There are separate Entry and Exit
- Barricade will be held open until the car passes the barricade.

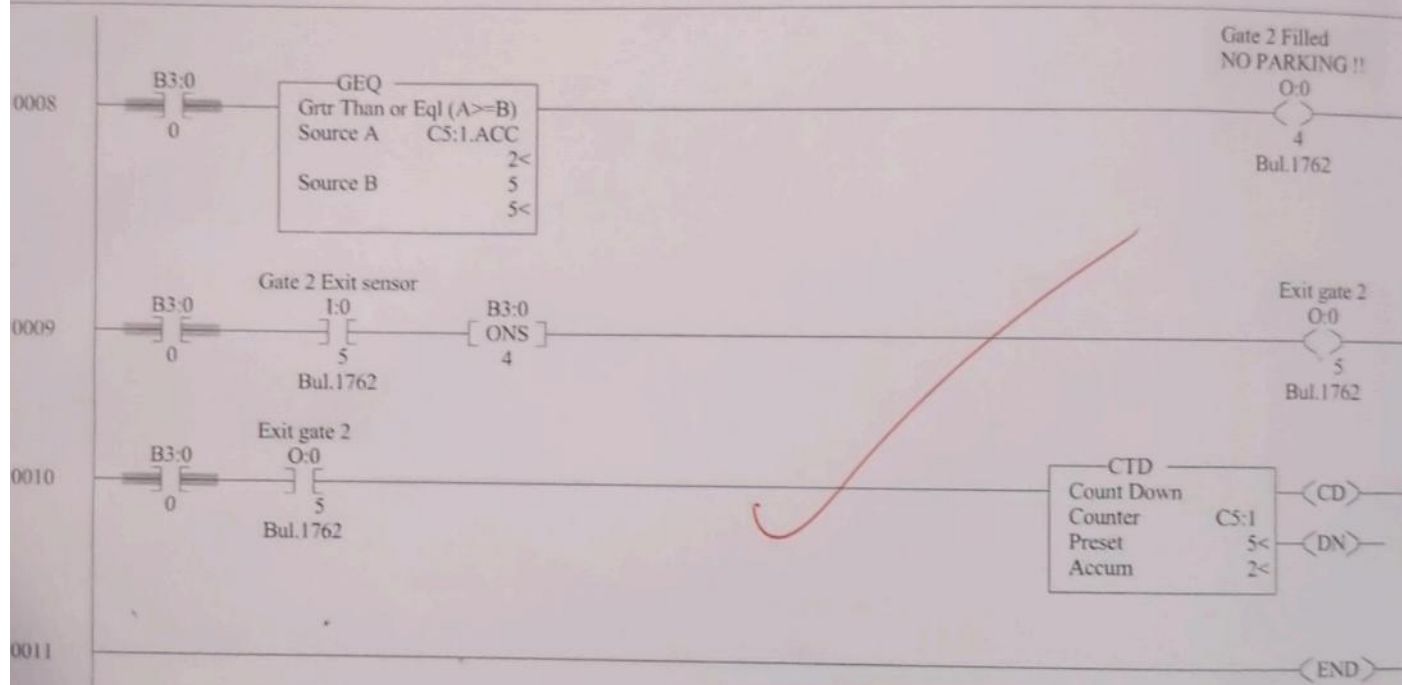
Observation -

- We newly used the counters and Timers as part of our Ladder Diagram.
- We used Counters to calculate the number of occupied places in the parking.
- We used Timers to hold up the barricade by checking the sensor.
- We also placed reset button.



CAR PARKING MARSHA

LAD 2 - --- Total Rungs in File = 12





Conclusion ~

The PLC based Car parking system was successfully implemented and executed. The system efficiently controls vehicle entry and exit while monitoring the available parking spaces automatically.

[Signature]
15/09